University of Florida, Department of Mathematics

Numerical Analysis PhD Exam, January 2025

Do 4 (four) of the first 5 (1-5) and 4 (four) of the last 5 problems (6-10).

1. Assume $A \in \mathbb{C}^{m \times m}$.

(a) Show that A has a Schur decomposition.

(b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.

2. If $q_1, \dots, q_n$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$ with $m > n$, prove that the orthogonal projector onto V is QQ^*, where Q is the matrix whose columns are the q_j.

3. Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$ and $\operatorname{rank}(A) = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A.

(a) Using the modified Gram–Schmidt process, write out expressions for q_1, q_2, q_3, the first three columns of Q in the QR decomposition of A.

(b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2.

4. Let $\|\cdot\|$ be a subordinate (induced) matrix norm.

(a) If E is a $n \times n$ with $\|E\| < 1$, then show $I + E$ is nonsingular and

$$\|(I+E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

(b) If A is a $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\|\|E\| < 1$, then show $A + E$ is nonsingular and

$$\|(A+E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

5. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$. Let u_j denote column j of U.

(a) Suppose $\operatorname{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\operatorname{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\operatorname{Null}(A^*)$.

(b) Suppose A is full rank and $x \neq 0$. Let σ_i, $i = 1, \dots, n$, be the singular values of A. Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that it holds.

6. $g(x) = \frac{x}{2} + \frac{a}{2x}$ has $x = \sqrt{a}$ as a fixed point where $a > 0$. Suppose we use the fixed point method with a starting point $x_0 > \sqrt{a}$. Use the theory of the fixed point method to determine the condition on x_0 which guarantees

$$|x_{k+1} - \sqrt{a}| \leq \frac{1}{2}|x_k - \sqrt{a}|, \qquad k \geq 1.$$

7. Let $f \in C[0,1]$ and let $\sum_{i=1}^{n} w_i f(x_i)$ $(n \geq 1)$ be an approximation to $\int_0^1 f(x)\,dx$. Assume that $0 \leq w_i \leq 1$, $0 \leq x_i \leq 1$ $(i = 1, 2, \dots, n)$. Let

$$E_n(f) = \int_0^1 f(x)\,dx - \sum_{i=1}^{n} w_i f(x_i)$$

be the error in the approximation. Assume that $E_n(p_n) = 0$ for p_n, which denotes any polynomial of degree $\leq n$. Use the Weierstrass approximation theorem to prove that, given $\epsilon > 0$, there is an $N > 0$ such that $|E_n(f)| < \epsilon$ when $n > N$.

8. Given a differentiable function $f(x)$, consider the problem of finding a polynomial $p(x) \in \mathbb{P}^n$ such that

$$p(x_0) = f(x_0), \qquad p'(x_i) = f'(x_i), \qquad i = 1, 2, \dots, n,$$

where x_i, $i = 1, 2, \dots, n$, are distinct nodes. (It is not excluded that $x_1 = x_0$.) Show that the problem has a unique solution and describe how it can be obtained.

9. Let $\{p_i(x)\}_{i=0}^{\infty}$ be the sequence of orthogonal polynomials obtained from the Gram–Schmidt orthogonalization process of $\{x^i\}_{i=0}^{\infty}$. Consider the quadrature formula $Q(f) = \sum_{j=0}^{m} \alpha_j f(x_j)$ for $J(f) = \int_a^b f(x)\rho(x)\,dx$. Prove that $Q(f)$ is exact for polynomials of degree $\leq 2m+1$ if and only if the quadrature nodes x_j $(j = 0, 1, \dots, m)$ are the roots of $p_{m+1}(x)$ and the quadrature weights $\alpha_j = \int_a^b l_j(x)\rho(x)\,dx$, $(j = 0, 1, \dots, m)$.

10. Consider numerically solving the initial value problem

$$y'(t) = f(t, y), \qquad 0 < t \leq t_f, \qquad \text{with } y(0) = \eta.$$

Assume f is sufficiently differentiable and let h denote the step size. Show that the method

$$y_{n+2} - y_{n+1} = \frac{1}{4}h(f_{n+2} + 2f_{n+1} + f_n)$$

is A_0 stable.